

MTH 419 Introduction to Abstract Algebra Syllabus - Fall 2026

Lectures: Tue, Thu 11:00-12:20 AM, Math Building 150

Recitations: Fri 1:00-1:50 PM, Baldy 200G

Instructor: Bernard Badzioch

E-mail: badzioch@buffalo.edu

Office: Math Building 108

Office Hours: Thu 2:00-4:00 PM and by appointment.

TA: Michael Erickson

E-mail: erickso2@buffalo.edu

Office: TBD

Office Hours: TBD.

Course Resources:

- **Course Website:** mth419.ubmath.info. See this website for an up-to-date version of this syllabus, course lecture notes, homework assignments etc.
- **Lecture notes.** Skeletal lecture notes for each week will be posted on the course website. I will annotate these notes during lectures. Please print the notes and bring them to class to follow along.
- **Piazza.** We be using Piazza for course-related communications. The link to the Piazza webpage for this course is
<https://piazza.com/buffalo/fall2026/mth419/home>
If you have a question or comment related to the course please post it on Piazza. This will help other students who may deal the same issue. If you know the answer to a question somebody else posted on Piazza, please answer it. If you have a personal question (concerning your grade etc.), you can either send to me a private message on Piazza or contact me by e-mail.
- **Textbook.** Joseph A. Gallian, *Contemporary Abstract Algebra*.
This book is currently in the 11th edition, but any older edition is just as fine - they all cover the same material and changes between various editions are mostly cosmetic.
- **UBLearns.** UBLearns will be used to collect homework assignments.

Homework. Homework problems will be assigned every week. The goal of homework is to let you practice with the course material and to provide you with feedback on your work. With this mind, you will get back homework assignments with comments on your solutions, but each honest attempt at a solution will receive full credit - regardless if it is fully correct or not. The only rule is that you must fully understand solutions you are submitting. Occasionally, I may request to meet with you so you can explain

your work in person. If you will not be able to do it, you will receive no credit for the homework assignment, and additionally your final course grade will be lowered by one letter grade (from A to A-, from A- to B+ etc.) for each instance.

Weekly Digests. Each week you will be asked to submit a short (1-3 sentences) writeup on your study from the previous week. You will be also asked to submit a question (or questions) regarding the course. You can receive up to 10% credit for these write-ups. You can miss one such assignment without losing any credit, but your weekly digest credit will be lowered by 2% for each subsequent missed assignment (i.e. from 10% to 8% etc.).

Exams:

Midterm 1	Thursday, October 8	11:00-12:20 PM	Math Building 150
Midterm 2	Thursday, November 12	11:00-12:20 AM	Math Building 150
Final Exam	Thursday, December 10	11:45 AM-2:45 PM	Math Building 150

Grading:

Homework	20%
Digests	10%
Midterm 1	20%
Midterm 2	20%
Final Exam	30%

Learning outcomes. The goal of this course is to introduce basic notions and methods of abstract algebra. Course topics will include:

- Definition and elementary properties of groups.
- Cyclic groups, dihedral groups, symmetric groups.
- Homomorphisms and isomorphisms of groups.
- Subgroups
- Direct products of groups.
- Cosets, Lagrange theorem.
- Group actions on sets.
- Normal subgroups and quotient groups.
- Isomorphism theorems.
- Classification of finitely generated abelian groups.
- Rings.
- Ideals of rings and quotient rings.
- Ring homomorphisms.
- Integral domains.
- Polynomial rings.
- Fields.

Prerequisites. The main expectation is that students taking this course are familiar with reading and writing mathematical proofs to the extent that is covered in MTH 311. We will also use some notions related to sets and functions (also typically covered in MTH 311), as well as some elementary number

theory (divisibility of integers, the greatest common divisor etc.). From time to time we will also use linear algebra (matrices, determinants, vector spaces etc.). These topics are covered in MTH 309.

Math Help Center. The Math Help Center is an additional place (beside lectures, recitations, and office hours) where you can seek help with questions related to this course. The Math Help Center is located in room 110 of the Mathematics Building and it is staffed by math graduate students. You don't need an appointment to visit it, you can stop by at any time during its opening hours. The current schedule of the Math Help Center is posted here: www.buffalo.edu/cas/math/ug/help-center.html.

Incomplete grades. See the UB Catalog for the UB incomplete grades policy:
catalogs.buffalo.edu/content.php?catoid=11&navoid=571#incomplete-grades

Academic integrity. See the UB Catalog for the UB academic integrity policy:
catalogs.buffalo.edu/content.php?catoid=11&navoid=571#academic-integrity

Accessibility resources. If you need accommodations due to a physical or learning disability please contact the UB Accessibility Resources Office to make appropriate arrangements:
www.buffalo.edu/studentlife/who-we-are/departments/accessibility.html